
```
load('lin_sys.mat', 'C', 'sys');

Ts = 0.05; % Sampling time (in seconds)
sys_dt = c2d(sys, Ts, 'zoh'); % Discrete-time state-space model

Ad = sys_dt.A;
Bd = sys_dt.B;
Cd = C;

poles = eigs(Ad)
trans_zeros = tzero(sys_dt)
ctrb_mat = ctrb(Ad, Bd);
rank(ctrb_mat) % controllable

obsv_mat = obsv(Ad, Cd);
rank(obsv_mat) % observable

poles =

    1
    1
    1
    1
    1
    1
    1

trans_zeros =

   -0.9917
   -1.0000

ans =

    12

ans =

    12
```

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